Netflix Dataset Analysis Project Report

Internship Project 3 – Data Analyst (Intermediate Level)
Dataset: Netflix Titles Catalog (CSV)
Tools Used: Python (Pandas, NumPy, Matplotlib), Scikit-learn (K-Means, Silhouette Score)
Outputs: Cleaned dataset, EDA visualizations, insights, ML segmentation (clustering), and business conclusions

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1. Project Objective

The objective of this project is to perform a complete data analytics workflow on a Netflix dataset to generate meaningful business insights. The workflow includes:

- Loading and understanding the dataset
- Cleaning and preparing the data for analysis
- Performing Exploratory Data Analysis (EDA) using professional visuals
- Applying an ML model (K-Means clustering) to segment content into meaningful groups
- Extracting actionable insights and drawing conclusions for decision-making

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2. Dataset Overview

The Netflix dataset contains metadata about shows and movies available on Netflix.

2.1 Dataset Size and Structure

- **Total rows (after cleaning):** 8,787
- **Total columns (after feature engineering):** 15

2.2 Original Columns

- show_id – Unique identifier for each title
- type – Movie or TV Show
- title – Name of the title
- director – Director name
- country – Production country
- date_added – Date added to Netflix
- release_year – Year of release
- rating – Age rating (TV-MA, PG, etc.)

- duration – Duration in minutes or seasons
- listed_in – Genres/categories

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3. Step 1 – Data Loading (Python)

The dataset was loaded into Python using Pandas and then inspected for structure, datatypes, and basic quality checks.

Key Checks Performed

- Rows and columns count
- Column names verification
- Preview of first few records
- Data type inspection

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4. Step 2 – Data Cleaning (Python)

Even though the dataset had **0 missing values initially**, it contained placeholders such as "Not Given" which are not real values. Proper cleaning was needed to make the data analysis accurate.

4.1 Cleaning Actions Performed

A) Handling Missing Values

- "Not Given" values were converted to real missing values (**NaN**) for correct analysis.

✅ Missing values after cleaning:

- director: **2,587 missing**
- country: **287 missing**

Note: Missing values appeared only after cleaning because placeholders like “Not Given” were replaced with NaN.

B) Duplicate Removal

- Duplicate content records were removed by checking duplicates across all columns except show_id.
- This removed **3 duplicate entries**, reducing rows from **8,790 to 8,787**.

C) Data Type Correction

- date_added was converted into proper datetime format for time-series trend analysis.
- release_year was ensured as numeric.

D) Feature Engineering (New Columns Created)

To support better EDA and analysis, the following columns were created:

1. **duration_value**
   Extracted numeric duration value from text duration (e.g., "90 min" → 90, "2 Seasons" → 2)
2. **movie_minutes**
   Stores duration in minutes only for Movies
3. **tv_seasons**
   Stores number of seasons only for TV Shows
4. **added_year**
   Extracted year from date_added
5. **added_month**
   Extracted month from date_added

These engineered features were important for professional-level insights and charting.

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5. Step 3 – Exploratory Data Analysis (EDA)

EDA was performed to identify patterns, trends, and distributions using "WOW" visuals (bar charts, line charts, heatmaps, wordcloud, etc.).

5.1 Key EDA Areas Covered

- Content distribution (Movies vs TV Shows)
- Ratings distribution (target audience maturity)
- Top genres and category dominance
- Country-wise contribution and diversity
- Time trends: yearly and monthly additions
- Duration analysis (movies in minutes, shows in seasons)
- Content freshness (age of titles when added)

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6. Step 4 – Machine Learning Model (K-Means Clustering)

To enhance the project with ML, **K-Means Clustering** was used to segment Netflix titles into similar groups without using a target label (unsupervised learning).

6.1 Why Clustering?

Clustering helps discover hidden structure and create meaningful content segments, useful for:

- catalog grouping
- business segmentation
- recommendation strategy
- market content planning

6.2 Features Used in Clustering

The model used:

- Genres (listed_in) using TF-IDF
- rating
- country
- duration_value
- release_year
- added_year

6.3 Selecting the Best K

Different K values were tested and evaluated using **Silhouette Score**.

- ✅ **Best K selected: 3**
- ✅ **Silhouette Score: 0.2029**

Interpretation

Silhouette score indicates cluster separation quality. Since Netflix genres overlap heavily, moderate scores are expected and still meaningful for real-world catalog segmentation.

6.4 Cluster Results

Cluster counts:

- Cluster 0: **2,858**
- Cluster 1: **5,281**
- Cluster 2: **648**

6.5 Cluster Meaning (Based on Top Genres)

Cluster 0 – "International TV & Serialized Content Cluster"

Top genres:

- International TV Shows
- TV Dramas
- TV Comedies

Cluster 1 – "Mainstream International Movies Cluster (Core Catalog)"

Top genres:

- International Movies
- Dramas
- Comedies

Cluster 2 – "Niche Mixed Movies Cluster"

Top genres:

- Dramas
- Comedies
- International Movies
  (Separated due to different country/rating/year patterns, making it a niche segment)

6.6 Custom Input Feature (Model Prediction)

A user-friendly feature was added:

- User enters content info (genre, rating, country, duration, year, etc.)
- Model predicts which cluster it belongs to
- System also displays example titles from that cluster

This makes the ML model interactive and presentation-ready.

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7. Final EDA Insights (Numeric Insights)

1. Netflix catalog contains **8787 titles**, dominated by **Movies (6124)** compared to **TV Shows (2663)**.
2. Director information is missing for **2587 titles**, while country is missing for **287 titles** after converting 'Not Given' to NaN.

3. The most common rating is **TV-MA**, indicating Netflix strongly targets a mature audience segment.
4. Top genres include:
    - International Movies (**2752**)
    - Dramas (**2424**)
    - Comedies (**1673**)
    - International TV Shows (**1348**)
    - Documentaries (**869**)
5. Top contributing countries are:
    - United States (**3240**)
    - India (**1056**)
    - United Kingdom (**638**)
    - Pakistan (**420**)
    - Canada (**271**)
6. Netflix added maximum titles in **2019**, highlighting peak expansion.
7. Monthly seasonality shows highest additions in **July (Month 7)**, suggesting content release clustering.
8. Average movie duration is **~99.6 minutes**, indicating standard feature-length content.
9. Most TV shows have **1 season**, showing dominance of limited/short series.
10. Median content age at time of addition is **1 year**, meaning Netflix frequently adds recent content.

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8. Business Conclusions & Recommendations

Conclusion 1: Netflix is movie-heavy

Netflix has more movies than TV shows, indicating a broad catalog strategy aimed at diverse, quick-view content.

Conclusion 2: Mature audience focus

The dominance of TV-MA suggests the platform prioritizes mature content, which impacts marketing and recommendation design.

Conclusion 3: International content is Netflix's strength

Top genres and countries show strong international expansion and multi-region content sourcing.

Conclusion 4: Strategic growth period (2019)

The platform showed maximum expansion in 2019. Understanding such growth years supports content acquisition forecasting.

Conclusion 5: Content segmentation improves decision-making

The K-Means model created 3 meaningful content clusters which can help Netflix:

- Segment content library
- Improve content planning
- Support targeted promotions
- Group content for recommendation strategies

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9. Final Deliverables

- ✅ Cleaned dataset: netflix_final_cleaned.csv
- ✅ Python notebook with full workflow (loading → cleaning → EDA → clustering): https://colab.research.google.com/drive/1-heIP1XsF0icWoLOC_gTJkIbc7faOWdM
- ✅ ML clustering model + custom cluster prediction feature
- ✅ Insights and conclusions report (this document)